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Test Pattern

ALLEN CAREER INSTITUTE

path to success

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

NEET(UG)
MINOR
04-03-2023

ENTHUSIAST ADVANCE COURSE PHASE : MED-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to **fill 0012345678.**

This Booklet contains 28 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet, fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 Hours 20 Minutes** duration and this Test Booklet contains **200** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. In this Test Paper, each subject will consist of **two sections**. **Section A** will consist of **35** questions (all questions are mandatory) and **Section B** will have **15** questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.
4. In case of more than one option correct in any question, the best correct option will be considered as answer.
5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) ____________________

Form Number : in figures ____________________

: in words ____________________

Centre of Examination (in Capitals) : ____________________

Candidate's Signature : ____________________ Invigilator's Signature : ____________________

Your Target is to secure Good Rank in Pre-Medical 2024

SUBJECT : PHYSICS

Topic : GRAVITATION, NLM, FRICTION, ELECTROSTATICS (TILL DATE)

SECTION-A

Attempt All 35 questions

1. The time period of moon's revolution around the earth is approximately 29 days. If moon's mass were 2 fold its present value and all other things remain unchanged, the time period of moon's revolution would be nearly :-

(1) $29\sqrt{2}$ days (2) $\frac{29}{\sqrt{2}}$ days

(3) 29×2 days (4) 29 days

2. A body is projected up from the surface of the earth with a velocity equal to $\frac{3}{4}$th of escape velocity for earth's surface. If R be the radius of the earth, the maximum height it reaches is :-

(1) $\frac{3R}{10}$ (2) $\frac{9R}{7}$ (3) $\frac{8R}{5}$ (4) $\frac{9R}{5}$

3. The satellite of mass 'm' revolving in a circular orbit of radius 'r' around the earth has kinetic energy E. then its angular momentum about the center of earth is :-

(1) $\sqrt{\frac{E}{mr^2}}$ (2) $\frac{E}{2mr^2}$

(3) $\sqrt{2Emr^2}$ (4) $\sqrt{2Emr}$

4. Three forces are acting on a particle of mass 'm' initially in equilibrium. If the first two forces (R_1 and R_2) are perpendicular to each other and suddenly the third force (R_3) is removed then the acceleration of the particle is:-

(1) $\frac{R_3}{m}$ (2) $\frac{R_1+R_2}{m}$

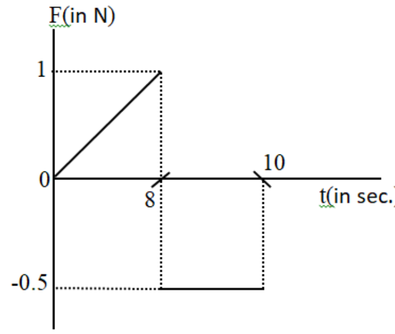
(3) $\frac{R_1-R_2}{m}$ (4) $\frac{R_1}{m}$

5. A toy of mass 1kg is placed on a spring balance. Suddenly, the toy jumps upwards. When it happens, the reading of the spring balance changes from 1 kg to 1.10 kg. what will be the value of the upward acceleration of the toy ? (g=10 m/s^2)

(1) 1 m/s^2 (2) 0.1 m/s^2

(3) 0.01 m/s^2 (4) 0.001 m/s^2

6.

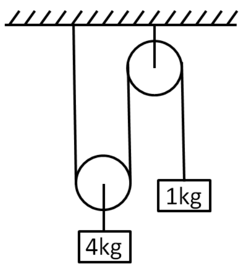


A force - time graph for the motion of a body is shown in the figure. The change in the momentum of the body between 0 to 10 sec. is -

(1) Zero (2) $4\frac{kg-m}{s}$

(3) $5\frac{kg-m}{s}$ (4) $3\frac{kg-m}{s}$

7.



In the system shown in the adjoining figure, the acceleration of the 1kg mass is :-

(1) $\frac{g}{4}$ downwards (2) $\frac{g}{4}$ upwards

(3) $\frac{g}{2}$ downwards (4) $\frac{g}{2}$ upwards

8. A body of mass 'm' is lifted up from earth's surface to a height 3 times the radius of earth. Change in potential energy of the body is : ('g' is acceleration due to gravity at earth's surface)

(1) 3mgR (2) $\frac{3}{4}$mgR

(3) $\frac{1}{3}$mgR (4) $\frac{2}{3}$mgR

9. A tunnel is dug along the diameter of the earth (radius R and mass M). There is a particle of mass 'm' at the centre of the tunnel. The minimum velocity given to the particle so that it just reaches to the surface of the earth is :-

(1) $\sqrt{\frac{GM}{R}}$

(2) $\sqrt{\frac{GM}{2R}}$

(3) $\sqrt{\frac{2GM}{R}}$

(4) It will reach with the help of a negligible velocity

10. Two satellites A and B of masses m_1 and m_2 ($m_1 = 2m_2$) are moving in circular orbits of radii r_1 and r_2 ($r_1 = 4r_2$) respectively, around the earth. If their time period are T_A and T_B, then the ratio T_A/T_B is :-

(1) 4 (2) 8 (3) 2 (4) 16

11. The radius of the earth is R. The height of a point vertically above the earth's surface at which acceleration due to gravity becomes 1% of its value at the surface is:-

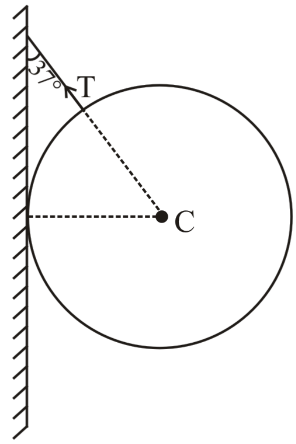
(1) 8 R (2) 9 R

(3) 10 R (4) 20 R

12. A uniform thick rope of length 5m is kept on frictionless surface and a force of 5N is applied to one of its end. Find tension in the rope at 1m from this end :

(1) 1 N (2) 3 N (3) 4 N (4) 5 N

13. A sphere of mass 2 kg is connected by using string as shown in figure. Find tension in string ($g = 10\ m/s^2$) :-



(1) 20 N (2) 15 N (3) 25 N (4) 30 N

14. The apparent weight of a person inside a lift is W_1 when lift moves up with a certain acceleration and is W_2 when lift move down with same acceleration. The weight of the person when lift moves up with constant speed is:-

(1) $\frac{W_1 + W_2}{2}$ (2) $\frac{W_1 - W_2}{2}$

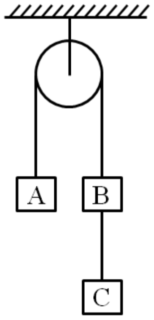
(3) $2W_1$ (4) $2W_2$

15. A body of mass 2 kg is sliding with a constant velocity of 4 m/s on a frictionless horizontal table. The force required to keep the body moving with the same velocity is:

(1) 8N (2) Zero

(3) 2×4^2N (4) (1/2) N

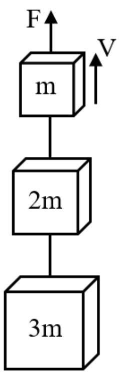
16. Three blocks A, B and C of same mass 2 Kg each are hanging as shown in the figure. The tension in the string connecting block B and C is :–



(1) Zero (2) 13.3

(3) 3.3 N (4) 19.6 N

17. Three blocks with masses m, 2m and 3m are connected by strings as shown in the figure, After an upward force F is applied on block m, the masses move upward at constant speed v. What is the net force on the block of mass 2m ? (g is the acceleration due to gravity)



(1) Zero (2) 2 mg

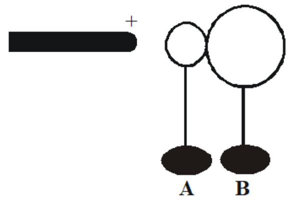
(3) 3 mg (4) 6 mg

18. A conductor has 14.4×10^{-19} C positive charge the conductor has :-

(1) 9 electron in excess (2) 27 electron in excess

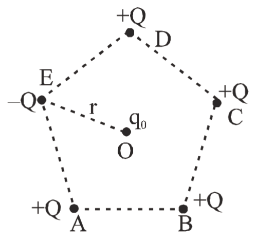
(3) 9 electron in short (4) 27 electron in short

19. Two uncharged, conducting spheres, A and B, are held at rest on insulating stands and are in contact. A positively charged rod is brought near sphere A as suggested in the figure. While the rod is in place, someone moves sphere B away from A. How will the spheres be charged, if at all?



	Sphere A	Sphere B
(1)	positive	positive
(2)	positive	negative
(3)	negative	positive
(4)	negative	negative

20. Five charges of equal values placed at corners of a regular pentagon of side 'a'. The force on q_0 which is placed at centroid 'O' is :-



(1) zero (2) $\frac{2kQq_0}{r^2}$, along OE

(3) $\frac{2kQq_0}{r^2}$, along EO (4) $\frac{kQq_0}{r^2}$, along OE

21. Two particles A and B (B is right of A) having charges 8×10^{-6} C and -2×10^{-6} C, respectively, are held fixed with separation of 20 cm. Where should a third charged particle be placed so that it does not experience a net electric force :-

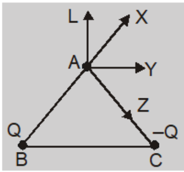
(1) 5 cm right of B (2) 5 cm left of A

(3) 20 cm left of A (4) 20 cm right of B

22. Two small identical spheres, each of mass 1 g and carrying same charge 10^{-9} C are suspended by threads of equal lengths. If the distance between the centres of the spheres is 0.3 cm in equilibrium then the inclination of the thread with the vertical will be :-

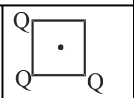
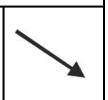
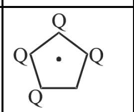
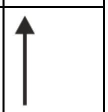
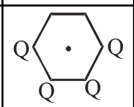
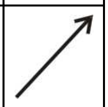
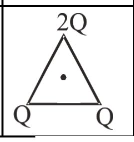
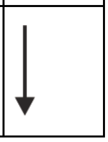
(1) $\tan^{-1}(0.1)$ (2) $\tan^{-1}(2)$
(3) $\tan^{-1}(1.5)$ (4) $\tan^{-1}(0.6)$

23. For the given figure the direction of electric field at A will be :



(1) towards AL (2) towards AY
(3) towards AX (4) towards AZ

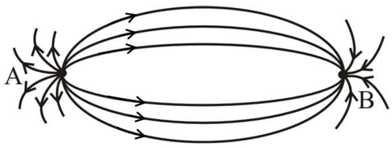
24. In the given diagrams the direction of electric field at point O is given in list-II (O is circumcenter of the given regular polygon). charge Q is positive. Match the direction of electric field for the given arrangement :-

List -I		List -II	
(P)		(1)	
(Q)		(2)	
(R)		(3)	
(S)		(4)	

Code :-

	P	Q	R	S
(1)	2	1	4	3
(2)	3	1	2	4
(3)	3	2	1	4
(4)	1	4	3	2

25. The spatial distribution of the electric field due to two charges (A and B) is shown in figure. Which one of the following statements is correct ?

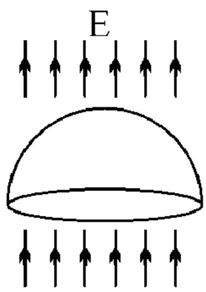


(1) A is +ve and B is -Ve and |A| > |B|.
(2) A is -Ve and B is +Ve and |A| = |B|.
(3) Both are +Ve but A > B.
(4) Both are -Ve but A > B.

26. If an electric field is given by $\left(10\hat{i}+3\hat{j}+4\hat{k}\right)\cdot\frac{N}{C}$, then calculate the electric flux through a surface of area 10 units lying in y-z plane -

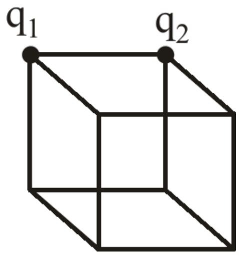
(1) 100 units (2) 10 units
(3) 30 units (4) 40 units

27. A hemispherical surface (half of a spherical surface) of radius R is located in a uniform electric field E that is parallel to the axis of the hemisphere. What is the magnitude of the outgoing electric flux through the hemisphere surface ?



(1) 0
(2) $4\pi R^2E/3$
(3) $2\pi R^2E$
(4) πR^2E

28. Flux through the cube is -



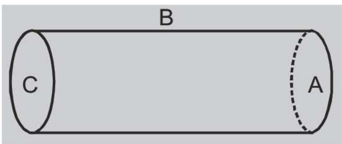
(1) $\frac{q_1}{8\epsilon_0}$

(2) $\frac{q_2}{8\epsilon_0}$

(3) $\frac{q_1 + q_2}{8\epsilon_0}$

(4) $\frac{q_1 - q_2}{8\epsilon_0}$

29. A hollow cylinder has a charge q coulombs located within it symmetrically. If ϕ is the electric flux in units of volt-meter associated with the curved surface B, the flux linked with the plane surface A in units of volt-meters will be :-



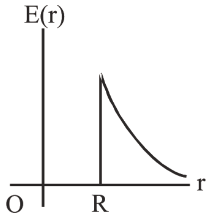
(1) $\frac{q}{\epsilon_0} - \phi$

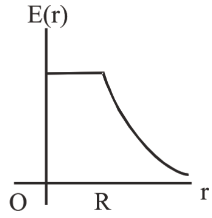
(2) $\frac{1}{2}\left[\frac{q}{\epsilon_0} - \phi\right]$

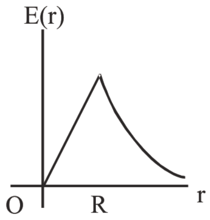
(3) $\frac{q}{2\epsilon_0}$

(4) $\frac{\phi}{3}$

30. A uniform sphere of radius R has a charge Q spread uniformly over its volume. Which of the following graphs most closely represents the electric field E(r) produced in the range $0 \leqslant r < \infty$, where r is the distance from the centre of the sphere?

(1)


(2)


(3)


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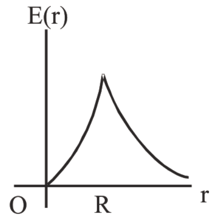


31. The electric field at a distance $\frac{3R}{2}$ from the centre of a charged conducting spherical shell of radius R is E. The electric field at a distance $\frac{R}{2}$ from the centre of the sphere is :-

(1) zero

(2) E

(3) E/2

(4) E/3

32. Two point charges +8q and –2q are located at x = 0 and x = 2L respectively. The location of the point on the x-axis at which the net electric field due to these two point charge is zero is:

(1) 8L

(2) 4L

(3) 2L

(4) $\frac{L}{4}$

33. Two charges, each equal to q, are kept at x = –a and x = a on the x-axis. A particle of mass m and charge $q = \frac{q_0}{2}$ is placed at the origin. If charge q_0 is given a small displacement (y <<a) along the y-axis, then the resulting motion of the particle is such that it will :

(1) execute SHM about the origin

(2) Oscillate but not execute SHM

(3) move towards origin and will become stationary

(4) Not return back to the equilibrium position

34. **Statement-I :** Charge is quantized.
Statement-II : Charge which is less than 1C is not possible.

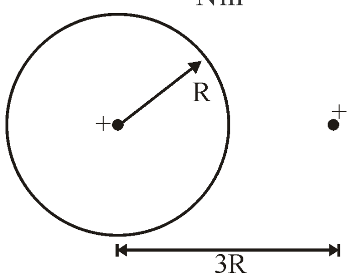
(1) Both statement I and II are true

(2) Both statement I and II are false

(3) Statement I is true but II is false

(4) Statement II is true but I is false

35. Find the flux of the electric field through a spherical surface of radius R due to a charge of 8.85×10^{-8} C at the centre and another equal charge at a point 3R away from the centre. (Given $\varepsilon_0 = 8.85 \times 10^{-12} \frac{C}{Nm^2}$)



(1) 10^4 (V-m)
(2) 10^{-4} (V-m)
(3) 10^{-8} (V-m)
(4) None

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

36. Let the minimum external work done in shifting a particle from center of earth to earth's surface be w_1 and that from surface of earth to infinity be w_2 . Then $\frac{w_1}{w_2}$ is equal to :-

(1) 1:1
(2) 1:2
(3) 2:1
(4) 1:3

37. Acceleration due to gravity on moon is $\frac{1}{6}$th of the acceleration due to gravity on earth. If the ratio of densities of earth (ρ_e) and moon (ρ_m) is $\frac{5}{3}$ (i.e. $\frac{\rho_e}{\rho_m} = \frac{5}{3}$) then radius of moon ($R_m$) in terms of R_e will be :-

(1) $\frac{5}{18} R_e$

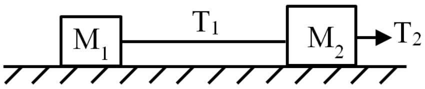
(2) $\frac{1}{6} R_e$

(3) $\frac{7}{18} R_e$

(4) $-\frac{1}{2\sqrt{3}} R_e$

38. How much deep inside the earth (radius R) should a man go, so that his weight becomes one fourth of that on the earth's surface ?

(1) $\frac{R}{4}$
(2) $\frac{R}{2}$
(3) $\frac{3R}{4}$
(4) None of these

39. 

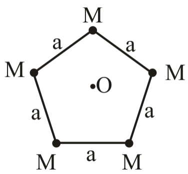
Two masses M_1 and M_2 are accelerated uniformly on a frictionless surface as shown in figure. The ratio of the tensions $\frac{T_1}{T_2}$ is :-

(1) $\frac{M_1}{M_2}$ (2) $\frac{M_2}{M_1}$

(3) $\frac{M_1+M_2}{M_2}$ (4) $\frac{M_1}{M_1+M_2}$

40. A body of mass 5 kg starts from the origin with an initial velocity $\vec{u}=(30\hat{i}+40\hat{j})$ m/s. if a constant force $\vec{F}=-(\hat{i}+5\hat{j})$ N acts on the body, the time in which the y component of the velocity becomes zero is :-

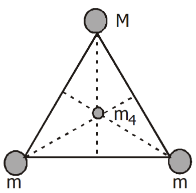
(1) 5 sec. (2) 20 sec. (3) 40 sec. (4) 80 sec.

41. 

Find net gravitational field intensity at O.

(1) $\frac{GM}{a^2}$ (2) $\frac{2GM}{a^2}$ (3) 0 (4) $\frac{2\sqrt{3}GM}{a^2}$

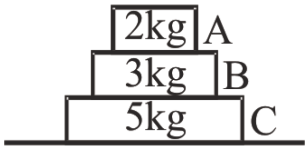
42. Two spheres of mass m and a third sphere of mass M form an equilateral triangle, and a fourth sphere of mass m_4 is at the center of the triangle. If the net gravitational force on m_4 from the three other spheres is zero, then :



(1) $M = m$ (2) $m = m/\sqrt{3}$

(3) $M = \sqrt{3}m$ (4) $M = \sqrt{2}m$

43. If all the surfaces are smooth then find force exerted by C on B :-



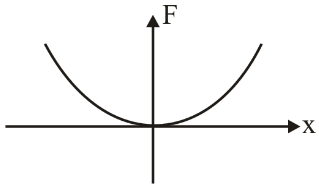
(1) 30 N

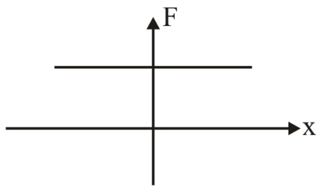
(2) 50 N

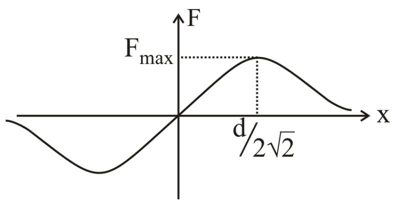
(3) 20 N

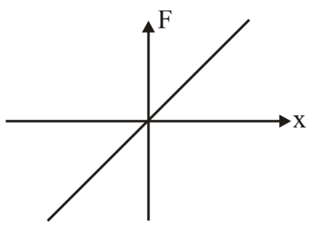
(4) 100 N

44. Two point charge Q, are fixed at $y=\frac{d}{2}$ and $y=-\frac{d}{2}$. A charge q placed on x axis variation of electric force experiment by q v/s x

(1) 

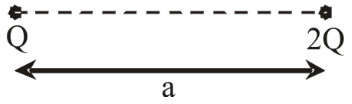
(2) 

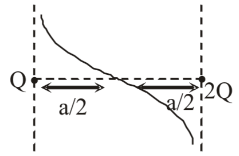
(3) 

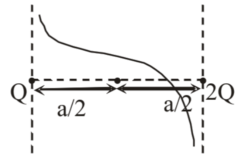
(4) 

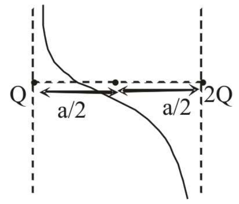
45. Find the variation of electric field on the line joining the two charges -

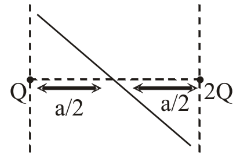
[Take +x axis as +ve direction of $\vec{E}$]



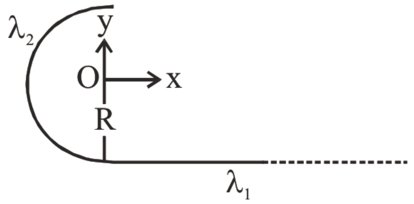
(1) 

(2) 

(3) 

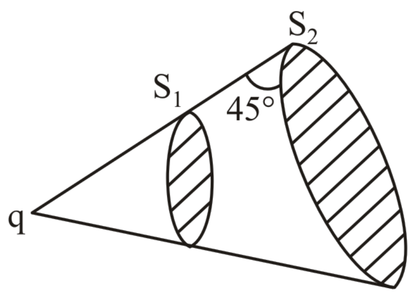
(4) 

46. In the figure shown, find the ratio of the linear charge densities λ_1 (on semi-infinite straight wire) and λ_2 (on semi-circular part) that is λ_1/λ_2 so that the field at O is along y direction.



(1) 2 (2) 1.5 (3) 3 (4) 2.5

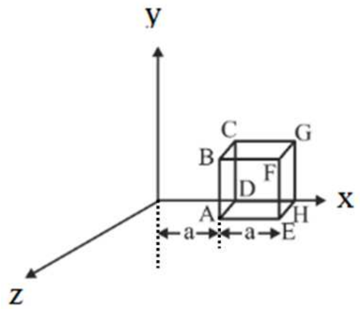
47. In the given figure flux through surface S_1 is ϕ_1 & through S_2 is ϕ_2 which of the following is correct ?



(1) $\phi_1 = \phi_2$ (2) $\phi_1 > \phi_2$

(3) $\phi_1 < \phi_2$ (4) None

48. A cube of side a is placed in electric field given by $\vec{E} = b\sqrt{x}\,\hat{i}$ (b = constant). Flux linked with cube is :-



(1) $ba^{1/2}(1-\sqrt{2})$ (2) $ba^{7/2}(1-\sqrt{2})$

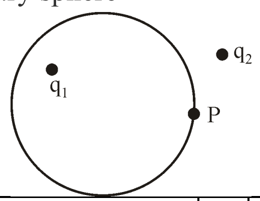
(3) $ba^{3/2}(\sqrt{2}-1)$ (4) $ba^{5/2}(\sqrt{2}-1)$

49. Four charges equal to –Q each are placed at the four corners of a square and a charge q is at its centre. If the system is in equilibrium, the value of q is :

(1) $-\frac{Q}{4}(1+2\sqrt{2})$ (2) $\frac{Q}{4}(1+2\sqrt{2})$

(3) $-\frac{Q}{2}(1+2\sqrt{2})$ (4) $\frac{Q}{2}(1+2\sqrt{2})$

50. In the figure shown P is a point on the surface of an imaginary sphere



	Column-I		Column-II
(A)	Electric field at point P	(p)	due to q_1 only
(B)	Electric flux through a small area at P	(q)	due to q_2 only
(C)	Electric flux through whole sphere	(r)	due to both q_1 and q_2

(1) A → q; B → p; C → p

(2) A → r; B → r; C → p

(3) A → r; B → p; C → q

(4) A → q; B → r; C → p

SUBJECT : CHEMISTRY

Topic : CHEMICAL KINETIC (COMPLETE), SOLUTIONS (TILL CONC. TERMS), CLASSIFICATION AND NOMENCLATURE, STRUCTURAL ISOMERISM, GEOMETRICAL ISOMERISM

SECTION-A

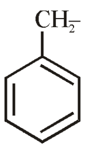
Attempt All 35 questions

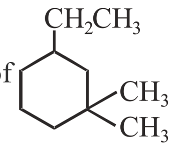
51. Which of the following match is incorrect?

(1) $CH_3-\underset{CH_3}{\overset{CH_3}{C}}-$: tert-butyl

(2) $CH_2=CH-CN$: Vinyl cyanide

(3) $CH_3-CH_2-\underset{CH_3}{CH}-OH$: Isobutyl alcohol

(4)  : Benzyl

52. IUPAC name of  is :-

(1) 1,1-Dimethyl-3-methylcyclohexane

(2) 3-Ethyl-1,1-dimethylcyclohexane

(3) 3,3-Dimethyl-1-ethylcyclohexane

(4) 1-Ethyl-3,3-dimethylcyclohexane

53. Which of the following represent pair of chain isomer :-

(1) n-pentane, Iso pentane

(2) n-pentane, Neo pentane

(3) Iso pentane, Neo pentane

(4) All

54. The correct IUPAC name of isooctane is :-

(1) 2,4,4–Trimethyl pentane

(2) 2,2,4–Trimethyl pentane

(3) 2,2,4–Trimethyl butane

(4) 2,4–Dimethyl pentane

55. Correct IUPAC name of compound is :-

$$CH_3-\underset{Br}{CH}-\overset{O}{\overset{\|}{C}}-O-C_2H_5$$

(1) 2-Bromo-1-ethyl propanoate

(2) 1-Ethyl-2-bromopropanoate

(3) Ethyl-2-bromopropanoate

(4) Ethyl-3-bromo propanoate

56. The IUPAC name of Compound

$$Cl-\underset{Cl}{\overset{Cl}{C}}-CH_2-CHO$$:-

(1) 2,2,2-Trichloropropanal

(2) 1,1,1-Trichloropropanal

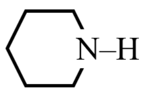
(3) 3,3,3-Trichloropropanal

(4) 1,2,1-Dichloromethanal

57. Which is tertiary amine ?

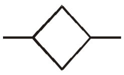
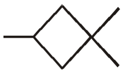
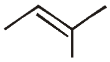
(1) $CH_3-\underset{CH_3}{\overset{CH_3}{C}}-NH_2$

(2) $CH_3-NH-\underset{CH_3}{\overset{CH_3}{C}}-CH_3$

(3) 

(4) $CH_3-\underset{CH_3}{N}-CH_2-CH_3$

58. Which of the following compounds show geometrical isomerism ?


(I)

(II)

(III)

(IV)

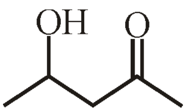
(1) I and II (2) II and III
(3) I and III (4) All

59. IUPAC Name of Neopentane is:-

(1) 2, 2-Dimethyl propane
(2) 2-Methyl propane
(3) 2, 2-Dimethyl butane
(4) 2-Methyl Butane.

60. Which of the following isomerism is not-shown by alkene :

(1) Chain isomerism
(2) Position isomerism
(3) Ring-chain isomerism
(4) Metamerism

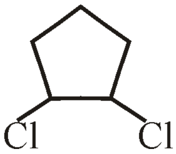
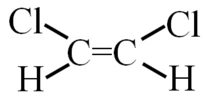
61. IUPAC Name of  is :-

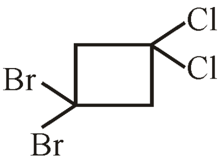
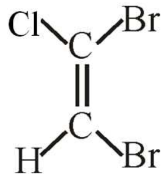
(1) 2-Hydroxy pentan-4-one
(2) 4-Hydroxy pentan-2-one
(3) 2-Oxo pentan-4-ol
(4) 4-Hydroxy-2-oxo pentane

62. The number of σ and π bond in isobutylene $\left[CH_3–\underset{\underset{CH_3}{|}}{C}=CH_2\right]$

(1) 11, 1 (2) 10, 1
(3) 9, 1 (4) 12, 1

63. Which of the following can not show geometrical isomerism ?

(1)  (2) 

(3)  (4) 

64. What is the hybridization of C_1, C_2, C_3 carbon in the following compound ?

$$CH_3–\underset{\underset{1}{\downarrow}}{CH_2}–\underset{\underset{2}{\downarrow}}{CH}=CH–\underset{\underset{3}{\downarrow}}{C}\equiv C–H$$

(1) sp^3, sp^3, sp^3 (2) sp^3, sp^2, sp^2
(3) sp^3, sp^2, sp (4) sp, sp^2, sp^2

65. Common name of $CH_3–O–C_2H_5$ is :

(1) Ethyl methyl ether
(2) Methyl ethyl ether
(3) Methoxy ethane
(4) Ethoxy methane

66. Which of the following is a example of functional group isomer.

(1) $CH_3–\underset{\underset{O}{\|}}{C}–CH_3,\ CH_3–CHO$
(2) $CH_3–CH_2–OH,\ CH_3–CHO$
(3) $CH_3–CH_2–CHO,\ CH_3–\underset{\underset{O}{\|}}{C}–CH_3$
(4) $CH_3–\underset{\underset{O}{\|}}{C}–CH_3,\ CH_3–CH_2–OH$

67. IUPAC name of compound

$$\begin{array}{l} CH_3-CH-CH-CH_3 \\ \quad\;\;\; | \qquad\; | \\ \quad\;\; CH_3 \;\; C_2H_5 \end{array}$$ is :

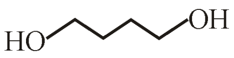
(1) 2-Methyl-3-ethyl butane

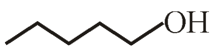
(2) 2-Ethyl-3-methyl butane

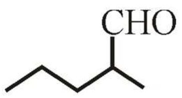
(3) 2, 3-Dimethyl Pentane

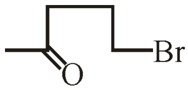
(4) 3, 4-Dimethyl pentane

68. Which of the following compound has 4 carbon atom in the parent chain ?

(1) 

(2) 

(3) 

(4) 

69. For a given reaction A $\rightarrow$ Product, rate is 1×10^{-4} Ms^{-1} when [A] = 0.01 M and rate is 1.41×10^{-4} Ms^{-1} when [A] = 0.02 M. Hence, rate law is-

(1) $\frac{-d[A]}{dt} = k[A]^2$ (2) $\frac{-d[A]}{dt} = k[A]$

(3) $\frac{-d[A]}{dt} = \frac{k}{4}[A]$ (4) $\frac{-d[A]}{dt} = k[A]^{1/2}$

70. For the reaction, $2NO + Br_2 \rightarrow 2NOBr$, the following mechanism has been given,

$NO + Br_2 \underset{}{\overset{\text{fast}}{\rightleftharpoons}} NOBr_2$

$NOBr_2 + NO \xrightarrow{\text{slow}} 2NOBr$

Hence rate law is,

(1) $\frac{dx}{dt} = k\,[NO]^2\,[Br_2]$

(2) $\frac{dx}{dt} = k\,[NO]\,[Br_2]$

(3) $\frac{dx}{dt} = k\,[NOBr_2]\,[NO]$

(4) $\frac{dx}{dt} = k\,[Br_2]^2\,[NO]$

71. The energies of activation for forward & reverse reactions for $A_2 + B_2 \rightleftharpoons 2AB$ are 180 kJ mol^{-1} & 200 kJ mol^{-1} respectively. The presence of catalyst lower the Ea of both (forward & reverse) reactions by 100 kJ mol^{-1}. The enthalpy change of the reaction ($A_2 + B_2 \rightarrow 2AB$) in the presence of catalyst will be (in kJ mol^{-1})

(1) 300 (2) 120 (3) 280 (4) –20

72. The rate of law for the reaction below is given by the expression k[A] [B]

A + B $\rightarrow$ Product

If the concentration of B is increased from 0.1 to 0.3 mol, keeping the value of A at 0.1 mol, the rate constant will be-

(1) 3k (2) 9k (3) k (4) k/3

73. The activation energy of a chemical reaction can be determined by-

(1) Evaluating rate constant at two different temperature

(2) Changing the concentration of reactants

(3) Evaluating the concentration at two different temperatures of reactants

(4) Evaluating rate constant at standard temperature

74. The relationship between $t_{50\%}$ & $t_{90\%}$ for a zero order reaction is-

(1) $t_{90\%} = 1.8 \times t_{50\%}$ (2) $t_{90\%} = 2 \times t_{50\%}$

(3) $t_{90\%} = 9 \times t_{50\%}$ (4) $t_{90\%} = 5 \times t_{50\%}$

75. The rate expression for the reaction $A(g) + B(g) \rightarrow C(g)$ is $r = kC_A^2 C_B^{1/2}$. What changes in the initial concentration of A & B will cause the rate of reaction increases by a factor of 8 ?

(1) $C_A \times 2$; $C_B \times 2$ (2) $C_A \times 2$; $C_B \times 4$

(3) $C_A \times 1$; $C_B \times 4$ (4) $C_A \times 4$; $C_B \times 2$

76. 2% (w/v) NaOH solution is equivalent to :-

(1) 1 N (2) 0.5 N (3) 0.1 N (4) 0.04 N

77. For a reaction $A + 2B \longrightarrow 6C + 2D$, if rate of dissappearance of A is 2.6×10^{-2} Ms^{-1}. Then what will be the value of $\frac{-\Delta[B]}{\Delta t}$?

(1) 2.6×10^{-2} (2) 1.3×10^{-2}

(3) 5.2×10^{-2} (4) 6.5×10^{-3}

78. In the series reaction $A \xrightarrow{k_1} B \xrightarrow{k_2} C \xrightarrow{k_3} D$. If $k_1 > k_2 > k_3$ then the rate determing step of the reaction is :-

(1) $A \rightarrow B$ (2) $C \rightarrow D$

(3) $B \rightarrow C$ (4) Any step

79. Select the rate law that corresponds to the data shown for the reaction $A + B \rightarrow C$

Exp.	[A]	[B]	Rate
1.	0.012	0.035	0.10
2.	0.024	0.070	0.80
3.	0.024	0.035	0.10
4.	0.012	0.070	0.80

(1) Rate = $k[B]^3$ (2) Rate = $k[B]^4$

(3) Rate $k = [A]\,[B]^3$ (4) Rate = $k[A]^2[B]^2$

80. A zero order reaction has $k = 0.025\ Ms^{-1}$ for $A \rightarrow$ product the disappearance of A. What will be concentration of A after 15 sec. if the initial concentration is 0.6M ?

(1) 0.06 M (2) 0.225 M (3) 0.125 M (4) 0.5 M

81. What is the unit of rate constant for following rate law : $r = k[A]^{1/2}\,[B]^{3/2}\,[C]^{-1}$

(1) mol $litre^{-1}$ $time^{-1}$ (2) mol^{-1} L $time^{-1}$

(3) $time^{-1}$ (4) mol^{-2} L^2 $time^{-1}$

82. The half life of a first order reaction is 120 hour. How long will it take to consume 93.75% of the reactant ?

(1) 480 min. (2) 240 min.

(3) 480 sec. (4) 28800 min.

83. 2g NaOH is present in 300 ml aqueous solution express its concentration in terms of percentage strength ?

(1) 0.33% (2) 0.55% (3) 0.66% (4) 6.6%

84. Dissolving 120 g of urea (mol wt = 60) in 1000 g of water gave a solution of density 1.15 g mL^{-1}. The molarity of the solution is

(1) 1.78 M (2) 2.30 M

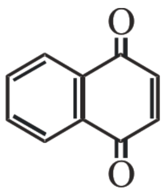
(3) 2.05 M (4) 2.22 M

85. P molal solution of a compound in benzene has mole fraction of solute equal to 0.2. The value of "P" is

(1) 2.0 (2) 13.88 (3) 14.3 (4) 3.2

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

86. Which type of hybridisation is present in naphthaquinone  is :-

(1) sp^3 (2) sp^2

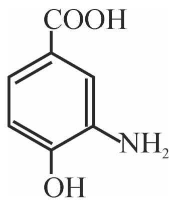
(3) sp (4) All of these

87.  &  Both are ?

(1) Chain isomer (2) Functional isomer

(3) Position isomer (4) Homologous

88. IUPAC name of following compound will be :



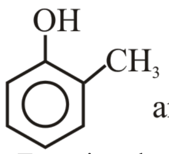
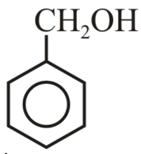
(1) p-Hydroxy m-amino benzoic acid

(2) 3-Amino-4-hydroxy benzoic acid

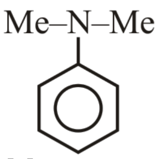
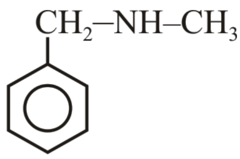
(3) 2-Amino-4-carboxy phenol

(4) 2-Hydroxy-5-carboxy aniline

89. Number of correct statement in following :

(a)  and 

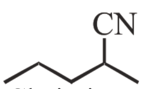
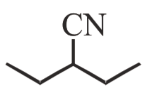
Functional group isomers

(b)  and 

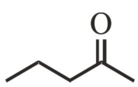
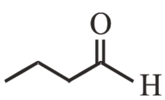
Metamers

(c) $CH_3–C{\equiv}C–CH_3$ and 

Ring chain isomer

(d)  and 

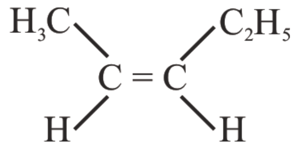
Chain isomers

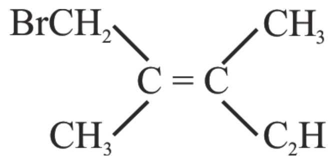
(e)  and 

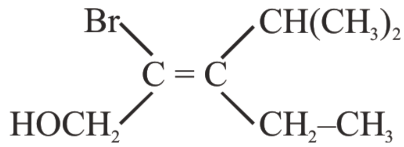
Functional group isomers

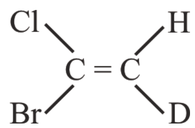
(1) 2 (2) 5 (3) 4 (4) 3

90. Which of the following has E-configuration ?

(1) 

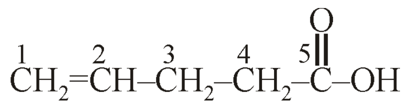
(2) 

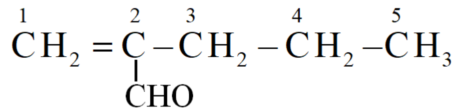
(3) 

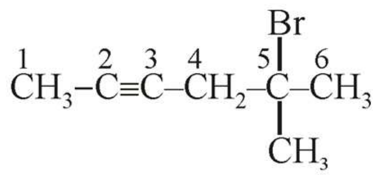
(4) 

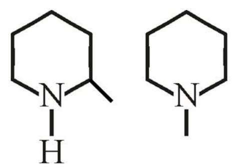
91. Which of the following has correct numbering according to IUPAC ?

(1) $\overset{1}{C}H_3–\overset{2}{C}{\equiv}\overset{3}{C}–\overset{4}{C}H_2–\overset{5}{C}H{=}\overset{6}{C}H_2$

(2) 

(3) 

(4) 

92.  are

(1) Identical

(2) Chain isomer

(3) Position isomer

(4) Functional group isomer

93. Which is not true for a 2^{nd} order reaction?

(1) It can have rate constant 1×10^{-2} L mol^{-1} s^{-1}

(2) Its half life is inversely proportional to its initial concentration

(3) Time to complete 75% reaction is twice of half life

(4) $T_{50}\% = \frac{1}{aK}$

94. A reaction involving two different reactant can never be

(1) 1st order reaction

(2) Unimolecular reaction

(3) Biomolecular reaction

(4) 2nd order reaction

95. The activation energy of a reaction A → B is 180 kJ mol^{-1}. By what factor the rate will increase if the temperature is increased from 27ºC to 37ºC?

(1) 1 (2) 10 (3) 100 (4) 1000

96. In the graph of Maxwell Boltzmann distribution of energy,
(i) Area under the curve must not change with increase in temperature.
(ii) Area under the curve increases with increase in temperature.
(iii) Area under the curve decreases with increase in temperature.
(iv) With increase in temperature curve broadens & shifts to the right hand side

(1) (ii) & (iv) (2) (iii) & (iv)

(3) (i) & (iv) (4) Only (iv)

97. Which of the following statement is incorrect about the collision theory of chemical reaction?

(1) It considers reacting molecules or atoms to be hard spheres & ignores their structural aspect

(2) Number of effective collisions determines the rate of reaction

(3) Collision of atoms or molecules possessing sufficient threshold energy results into the product formation

(4) Molecules should collide with sufficient threshold energy & proper orientation for the collision to be effective

98. For a reaction, activation energy is zero. What is the value of rate constant at 300 K, if $k = 1 \times 10^{-6}$ s^{-1} at 280K ($R = 8.314$ JK^{-1} mol^{-1}) :-

(1) 2×10^{-6} s^{-1}

(2) 4×10^{-6} s^{-1}

(3) 1×10^{-6} s^{-1}

(4) 1×10^{6} s^{-1}

99. The half life periods of a reaction at initial concentration 0.1 mol L^{-1} and 0.5 mol L^{-1} are 200 s and 40s respectively. The order of the reaction is :

(1) 1 (2) $\frac{1}{2}$

(3) 2 (4) 0

100. 6.8 g H_2O_2 is dissolved in 224 mL solution. This solution will be labelled as

(1) 0.224 V (2) 20 V

(3) 0.9 V (4) 10 V

SUBJECT : BIOLOGY-I

Topic : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH (TILL DATE), ORGANISM AND POPULATION, ECOSYSTEM, DOMESTICATION OF ANIMALS

SECTION-A

Attempt All 35 questions

101. In the structure of testis interstitial space contain :-

(1) Leydig cells

(2) Immuno competent cells

(3) Blood vessels

(4) All of the above

102. Epididymis located in :-

(1) Inside the testis

(2) Anterior surface of testis

(3) Posterior surface of testis

(4) Lateral surface of testis

103. Female accessory reproductive ducts includes :-

(1) Oviducts, Uterus, Ureter

(2) Oviducts, Vagina, Inguinal canal

(3) Oviducts, Uterus, Vagina

(4) Oviducts, Vagina, Ureter

104. Select incorrect statement regarding gametogenesis :-

(1) Spermatogonia are immature germ cells which meiotically divide and increase in number.

(2) 2 equal haploid secondary spermatocytes formed by meiotic division in primary spermatocyte.

(3) Spermatid head embedded in the sertoli cells during spermiogenesis.

(4) Release of sperm from seminiferous tubules is called spermiation.

105. Which of the following helps in motility & maturation of sperm :-

(1) Epididymis

(2) Vas deferens

(3) Accessory reproductive glands

(4) All of the above

106. Granulosa cells covering first surrounds :-

(1) Oogonia (2) Primary oocyte

(3) Primary follicle (4) Secondary oocyte

107. Presence of fluid filled cavity antrum is a characterstic of :-

(1) Secondary follicle (2) Primary follicle

(3) Tertiary follicle (4) Secondary oocyte

108. Primary oocyte complete first meiotic division in :-

(1) Sec. follicle (2) Graafian follicle

(3) Tertiary follicle (4) Primary follicle

109. Select the mismatched pair :-

(1) Limbs & digits - First trimester

(2) Major organ system - end of 12^{th} week

(3) First movement of foetus - 4^{th} month

(4) Eyelids forms & seperate - 2^{nd} trimester

110. In 1900 world population was :

(1) 2 billion (2) 1 billion

(3) 4 billion (4) 6 billion

111. Intense lactation inhibit or decrease level of :-

(1) Estrogen (2) LH

(3) Progesterone (4) Relaxin

112. **Assertion :-** Menstruation only occurs if the realsed ovum is not fertilized.

Reason :- Lack of menstruation may be indicative of pregnancy.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is true but the reason is false.

(4) Both Assertion & Reason are False.

113. If due to certain reason, mammary ampulla of breast is blocked in a lactating female, then milk would not be transferred from _(A)_ to _(B)_ in breast of this female.

(1) (A) Mammary tubule, (B) Mammary duct

(2) (A) Mammary duct, (B) Lactiferous duct.

(3) (A) Lactiferous duct, (B) Mammary duct.

(4) (A) Mammary duct, (B) Mammary tubule

114. Match the following homologous reproductive organs :-

	Male		Female
A.	Scrotum	i.	Glands of skene
B.	Penis	ii.	Labia majora
C.	Prostate	iii.	Bartholin's glands
D.	Cowper's gland	iv.	*Clitoris*

(1) A-ii, B-iv, C-i, D-iii

(2) A-ii, B-iv, C-iii, D-i

(3) A-iii, B-iv, C-i, D-ii

(4) A-ii, B-i, C-iv, D-iii

115. How many are part of external genitalia in female ?

Cervix, Hymen, Vagina, Clitoris, Labia minora

(1) 5

(2) 4

(3) 3

(4) 2

116. The first sign of the growing foetus may be noticed by listening to the heart sound carefully through the stethoscope embryo's heart is formed ________.

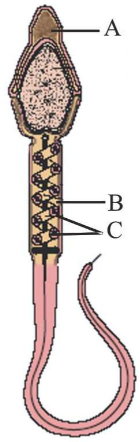
(1) By the end of the second month of pregnancy

(2) By the end of first trimester.

(3) After one month of pregnancy

(4) During fifth month

117.



Identify the A, B, C from above diagram. options :-

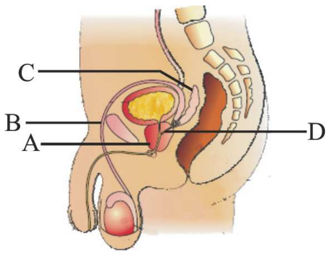
(1) A-Nucleus, B-Centriole, C-Mitochondria

(2) A-Acrosome, B-Middle piece, C-Mitochondria

(3) A-Plasma membrane, B-Mitochondria, C-Middle piece

(4) A-Plasma membrane, B-Mitochondria, C-Tail

118. In the given figure identify the part labelled as A, B, C & D :-



(1) A-vas deferences, B-seminal vesicle, C-prostate, D-ejaculatory duct

(2) A-prostate, B-vas deference, C-seminal vesicle, D-ejaculatory duct

(3) A-seminal vesicle, B-vas deferences, C-prostate, D-ejaculatory duct

(4) A-ejaculatory duct, B-prostate, C-vas deferences, D-seminal vesicle

119. Which of the following is first sign of pregnancy–

(1) PCOD (2) Dysmenorrhoea

(3) Amenorrhoea (4) Metrorrhagia

120. Inner side of seminiferous tubule is lined by :

(1) Spermatogonia and leydig cells

(2) Spermatid and leydig cells

(3) Spermatogonia and Sertoli cells

(4) P.G.C and white fibrous connective tissue

121. Which hormone level reaches peak during the luteal phase of menstrual cycle ?

(1) FSH (2) LH

(3) Estrogen (4) Progesterone

122. In primary oocyte of human female, Ist meiotic division is completed :-

(1) Prior to formation of yellow body

(2) Prior to ovulation

(3) Prior to birth

(4) Prior to puberty

123. Ejeculatory duct is formed by :-

(1) Urethra and Vas-defreus

(2) Urethra and duct of seminal vesicle

(3) Vas-defrens and duct of seminal vesicle

(4) Vas-defrens and epididymis

124. Which of the following is absent in placenta ?

(1) Endometrium (2) Chorion

(3) Amnion (4) Yolk sac

125. Select the number of sperms and ova produced by 100 primary spermatocytes and primary oocytes respectively.

(1) 400 and 400 (2) 200 and 200

(3) 400 and 100 (4) 200 and 50

126. A 25 yrs old female goes to doctor. She ask doctor that after 4 days there is due date of her menses and she wants to post-pone bleeding for some days as she will be engaged in some official work during those days what will be the doctor's suggestive treatment ?

(1) FSH and estrogen

(2) FSH and progesterone

(3) Progesterone only

(4) Estrogen only

127. Hymen membrane can be torn by
(a) First coitus
(b) Sudden fall or jerk
(c) Insertion of vaginal tampon
(d) Active participation in cycling & horse riding.
How many statements are correct.
(1) a, b and c (2) b, c and d
(3) a, b and d (4) a, b, c and d

128. How many of the following are not the part of fallopian tube ?
Isthmus, Cervix, Ampulla, Abdominal ostium, Infundibulum, Fundus
(1) Five (2) Four
(3) Three (4) Two

129. The kind of association between cattle egret birds and cattle ?
(1) Mutualism (2) Commensalism
(3) Parasitism (4) Predation

130. Which one of the following is an example of total stem parasite ?
(1) *Santalum* (2) *Raffflesia*
(3) *Viscum* (4) *Cuscuta*

131. Concept of resource partitioning is given by :-
(1) Gause (2) Conell's
(3) Mac Arthur (4) E. Mayer

132. Biological control method adopted in agriculture pest control are based on :-
(1) Competition (2) Amensalism
(3) Predation (4) Parasitism

133. Competition can be represented as :-
(1) Species A (–) ; species B (0)
(2) Species A (+) ; species B (+)
(3) Species A (–) ; species B (–)
(4) Species A (+) ; species B (0)

134. Primary succession does not take place on/in :-
(1) Bare rock
(2) Degraded forest
(3) Newly created pond
(4) Newly cooled lava

135. The species that invade a bare area are called :-
(1) Exotic species (2) Alien species
(3) Climax species (4) Pioneer species

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

136. Male accessory reproductive glands includes :-
(1) 2 seminal vesicle, 1 prostate, 2 bulbo urethral gland
(2) 1 seminal vesicle, 1 prostate, 2 bulbo urethral gland
(3) 2 seminal vesicle, 2 prostate, 1 cowper's gland
(4) 1 seminal vesicle, 2 prostate, 1 cowper's gland

137. Which of the following statement is correct regarding mammary glands.

(1) Presence of functional mammary gland is a characterstic of all mammals.

(2) The glandular tissue of each mammary gland is divided into 15-20 mammary tubules

(3) Mammary tubules fuse and form mammary duct these mammary ducts fuse and form lactiferous duct.

(4) Prolactin, estrogen, progesterone helps in the development of mammary gland.

138. How many primary follicles are left in female at the time of puberty :-

(1) 60,000 - 80,000 (2) 1,20,000 - 1,60,000

(3) 80,000 - 1,20,000 (4) 2,00,000 - 3,00,000

139. Which of the following statement is incorrect regarding menstrual cycle :-

(1) Breakedown of endometrial linings of uterus & its blood vessels is responsible for menstrual bleeding.

(2) Stress, poor health and other factors also responsible for lack of menstruation.

(3) Changes in the levels of pituitary and ovarian hermones induce changes in ovary and uterus.

(4) Level of LH is maximum during middle of menstural cycle while level of FSH is minimum during middle of menstrual cycle.

140. Perivitelline space present between :-

(1) Granulosa cells and corona radiata.

(2) Corona radiata and zona pellucida.

(3) Plasma membrane and vitelline membrane.

(4) Plasma membrane and zona pellucida.

141. Select incorrect statement :-

(1) Level of estrogen, Progesterone, Cortisol, Prolactin, thyroxin increase in foetal blood.

(2) HCG, HPL, Relaxin produced in female only during pregnancy.

(3) Differentiation starts in the cells of inner cell mass after implantation.

(4) Stem cells are present in inner cell mass and these cells have potency to give rise to all the tissues and organs.

142. **Assertion (A) :** Spermatogenesis starts at the age of puberty.

Reason (R) : There is significant rising in the secretion of gonadotropin releasing hormone (GnRH) at puberty.

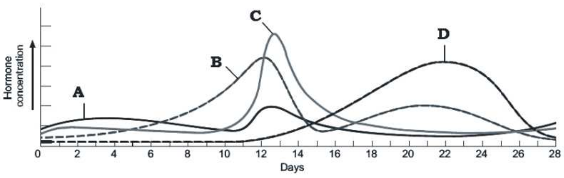
(1) Both Assertion (A) and Reason (R) are correct and (R) is correct explanation of (A).

(2) Both Assertion (A) and Reason (R) are correct and (R) is not correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) (A) is incorrect but (R) is correct.

143. Identify the hormones A, B, C and D from the graph by observing their concentrations in blood during the menstrual cycle of a female:



	A	B	C	D
(1)	FSH	Progesterone	Estrogen	LH
(2	FSH	Estrogen	LH	Progesterone
(3)	FSH	LH	Progesterone	Estrogen
(4)	LH	Estrogen	Progesterone	FSH

144. Consider the following four statements (a-d) and select five option which includes all the correct of ones only.

(a) The reproductive cycle in the female primates is called menstrual cycle

(b) Second polar body has 22 + X chromosome

(c) Antrum is found in secondary follicle

(d) For normal fertility, at least 60% sperms must have normal shape and size

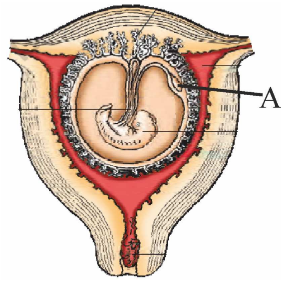
(1) a, b, c

(2) a, b, d

(3) a, b, c, d

(4) b, c, d

145. Observe the given diagram and choose correct option for labelled part :-



(1) Allantois – Less developed in human

(2) Amnion – Absent in human

(3) Yolk sac – Act as a haemopoietic organ

(4) Chorion – Release amniotic fluid

146. In spermatogenesis, first meiotic division leads to the formation of :-

(1) Spermatid

(2) Spermatozoa

(3) Primary spermatocyte

(4) Secondary spermatocyte

147. Identify the incorrectly matched option among the following :

(1) Sertoli cell – nutrition of developing sperms

(2) Penis – main copulatory organ of man

(3) Zona pellucida – prevention of ectopic pregnancy

(4) Hymen – a true sign of virginity

148. Human liver fluke (Trematode) intermediate hosts are :-

(1) Mosquito, Fish

(2) Snail, Fish

(3) Snail, Mosquito

(4) Sheep, Mosquito

149. Which of the following is a conduit for energy transfer across trophic levels ?

(1) Mutualism

(2) Parasitism

(3) Predation

(4) Proto cooperation

150. Select the incorrect match for the inter specific interaction :-

	Species-A	Species-B	Interaction
(1)	+	–	Parasitism
(2)	+	+	Mutualism
(3)	–	–	Competition
(4)	+	+	Commensalism

SUBJECT : BIOLOGY-II

Topic : HUMAN REPRODUCTION AND REPRODUCTIVE HEALTH (TILL DATE), ORGANISM AND POPULATION, ECOSYSTEM, DOMESTICATION OF ANIMALS

SECTION-A

Attempt All 35 questions

151. Find the odd one :-

(1) Epiphytes
(2) Barnacles - whale
(3) *Cuscuta* - Hedge plant
(4) Clown fish - sea anemone

152. **Assertion (A) :** Fig species can be pollinated only by its partner wasp species and no other species.
Reason (R) : In many species of fig trees, there is a tight one to one relationship (mutalism) with the pollinator species of wasp.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) (A) is correct but (R) is not correct.
(3) (A) is not correct but (R) is correct.
(4) Both (A) and (R) correct and (R) is the correct explanation of (A).

153. Relationship between *Yucca* plant flowers and *Pronuba* insects :

(1) Commensalism (2) Mutualism
(3) Amensalism (4) Parasitism

154. Which is not essential for self sustaining ecosystem ?

(1) Producers (2) Macro consumers
(3) Micro consumers (4) Decomposers

155. The principle of competitive release was stated by :-

(1) Mac Arthur (2) G.F. Gause
(3) Connell (4) C. Darwin

156. Which one of the following interactions is widely used in medical science for the production of antibiotics ?

(1) Comensalism (2) Mutualism
(3) Parasitism (4) Amensalism

157. Which one of the following pair of organism produce a large number of small sized offspring ?

(1) Pacific salmon fish, Bamboo
(2) Oysters, Pelagic fish
(3) Oysters, Pacific salmon fish
(4) Bamboo, Pelagic fish

158. The correct sequence of plants in a hydroscre is :-

(1) *Volvox* → *Pistia* → *Scirpus* → *Hydrilla* → Oak
(2) *Volvox* → *Vallisneria* → *Pistia* → *Scirpus* → Oak
(3) Oak → *Scirpus* → *Pistia* → *Vallisneria* → BGA
(4) BGA → *Scirpus* → *Pistia* → *Hydrilla* → Oak

159. The cuckoo (koel) lays its eggs in crow's nest known as :

(1) Hyper parasitism (2) Commensalism
(3) Mutualism (4) Brood parasitism

160. Which one of the following is an example of commensalism ?
(A) Orchid - Bee
(B) Orchid - Mango
(C) Clown fish - Sea Anemone
(D) Barnacles - Whale
(E) Sparrow - Seed

(1) A, B, C only (2) B, C, D
(3) A, B, E (4) C, D, E

161. Energy enters in a food chain though

(1) Producers (2) Decomposers

(3) Herbivores (4) Carnivores

162. The biomass available for consumption by the herbivores and the decomposers is called:

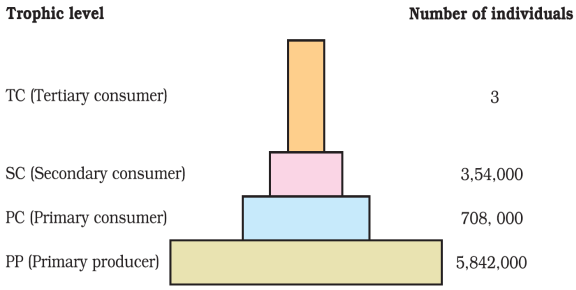
(1) net primary productivity

(2) secondary productivity

(3) standing state

(4) gross primary productivity

163.



The above graph represents :-

(1) Standing crop

(2) Standing state

(3) Biotic potential

(4) Lindeman's 10% law

164. Which one of the following is not a function of an ecosystem ?

(1) Decomposition (2) Productivity

(3) Stratification (4) Energy flow

165. The importance of ecosystem lies in -

(1) Flow of energy

(2) Cycling of materials

(3) Both the above

(4) None of the above

166. An assemblage of individual of different species living in the same habitat and having functional interaction with abiotic factors creates a physical structure called as

(1) Population (2) Biotic community

(3) Ecosystem (4) Landscape

167. Energy storage at consumer level is called :-

(1) Gross primary productivity

(2) Secondary productivity

(3) Net primary productivity

(4) Net community productivity

168. Read the following statements?

Statement (A) : Total amount of inorganic substance present in ecosystem is called standing crop.

Statement (B) : Total amount of living organic matter present in ecosystem is called standing state:

(1) Statements A correct and B incorrect

(2) Statements B correct and A incorrect

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

169. Amount of energy transferred from one trophic level to next is :

(1) 90 % (2) 10 % (3) 50 % (4) 1 %

170. According to the competitive exclusion principle, two species cannot continue to occupy the same :-

(1) Habitat

(2) Niche

(3) Territory

(4) Biome

171. The annual net primary productivity of the whole biosphere is approximately

(1) 50 billion tons (2) 170 billion tons

(3) 55 billion tons (4) 710 billion tons

172. Which statement is/are not correct ?

(1) Net primary productivity is available biomass for consumption to heterotrops

(2) Secondary productivity is defined as rate of formation of new organic matter by heterotrophs

(3) The annual net primary productivity of land is greater than ocean

(4) NCP increases during succession

173. Which one of the following types of organism occupy more than one trophic level in a pond ecosystem ?

(1) Frog (2) Phytoplankton

(3) Fish (4) Zooplankton

174. In grassland ecosystem the number of primary producer is about 6 million. The number of top carnivore supported by it would be :-

(1) 60 (2) 30 (3) 6 (4) 3

175. In the following statements choose correct options:-

(a) Wasp laying egg in a fig fruit is an example of mutual relationship

(b) Mycorrhiza is an example of ectoparasitism

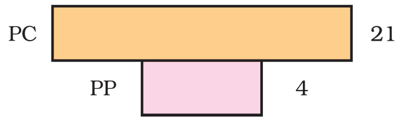
(c) Physiological ecology is ecology at organism level

(d) Competitive exclusion principle was given by Charls Elton

(1) a & c (2) a, c, d

(3) b and d (4) b, c and d

176. Above ecological pyramid shows :-



(1) Pyramid of number in aquatic ecosystem i.e. is upright

(2) Pyramid of biomass in terrestrial ecosystem i.e. is inverted

(3) Pyramid of energy

(4) Pyramid of biomass in aquatic ecosystem i.e. inverted

177. Given below is a food chain of terrestrial ecosystem:-

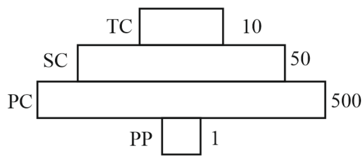
Grass → Grass hopper → Frog → Snake → Hawk

If the solar energy is 10,00000 J, what will be the amount of energy obtained by frog and hawk respectively ?

(1) 100 J, 1 J (2) 1000 J, 10 J

(3) 1000 J, 100 J (4) 10000 J, 100 J

178. Given below is an imaginary pyramid of numbers. What could be one of the possibilities about certain organisms at some of the different levels?



(1) Level one PP is "Pipal trees" and the level SC is "sheep"

(2) Leval PC is "rats" and level SC is "cats"

(3) Level PC is "insects" and level SC is "small insectivorous birds

(4) Level PP is "Phytoplanktons" in sea and "Whale" on top level TC.

179. **Assertion :** Artificial insemination helps us overcome several problems of normal matings.
Reason : In Multiple ovulation embryo transfer technology animal is either mated with an elite bull or artificially inseminated.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

180. What is correct about inbreeding ?

(1) Breeding of unrelated animals of same breed but having no common ancestor for 4-6 generations
(2) Mating between superior male of one breed with superior female of another breed
(3) Male and female of two different species are mated
(4) It increases homozygosity

181. Read the following four statements (a-d):
(a) Inbreeding increases homozygosity.
(b) Inbreeding is done between the animals of same breed.
(c) Inbreeding cannot be used to develop purelines in cattle.
(d) Inbreeding exposes harmful recessive genes.
How many statements are incorrect among these?

(1) One
(2) Two
(3) Three
(4) Four

182. Hisardale is a result of mating between:

(1) Bikaneri ewes and Marino rams
(2) Bikaneri ewes and Bikaneri rams
(3) Marino rams and Marino ewes
(4) Bikaneri rams and Marino ewes

183. The contribution of India and China to the world farm produce is :-

(1) 70%
(2) 25%
(3) 97%
(4) 11%

184. Which type of cross should be done between the animals to combine the qualities of two different breeds?

(1) Inbreeding
(2) Out-crossing
(3) Cross-breeding
(4) Interspecific hybridization

185. How many statements are correct about "MOET":-
(A) In this method, A cow is administered hormones with FSH like activity.
(B) The fertilized eggs transferred into surrogate mother when they achieve a stage of more than 32 cells.
(C) Genetic mother produces 6–8 ova per cycle in this method.

(1) Zero
(2) Two
(3) One
(4) Three

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 questions out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

186. Study the four statements (a-d) given below and select the two correct ones out of them :

(a) A lion eating a deer and a sparrow feeding on seed are ecologically similar in being predator.

(b) Predator star fish *Pisaster* helps in maintaining species diversity of some vertebrates.

(c) Predators ultimately lead to the extinction of prey species.

(d) Production of chemicals such as nicotine, strychine by the plants is a way of defence against predator.

The two correct statements are :-

(1) (a) and (b) (2) (b) and (c)

(3) (c) and (d) (4) (a) and (d)

187. Which one of the following is most appropriately defined ?

(1) Amensalism is a relationship in which one species is adversely affected where as the other organism stay unaffected.

(2) Predator is an organism that catches other organism.

(3) Parasite is an organism which always lives inside the body of other organism and may kill it.

(4) Host is an organism which provide only food to another organism.

188. Given below are two statements :

Statement-I : The entire sequence of communities that successively change in a given area are called sere.

Statement-II : In succession future seral communities can not be predicted as it is not a directional process.

(1) Both statements-I and II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but statement-II is correct.

(4) Both statements-I and II are correct.

189. Which one of the following has highest number of producers per unit area ?

(1) Forest (2) Grassland

(3) Desert (4) Pond

190. Which one of the following is an example of total root parasite ?

(1) *Santalum* (2) *Rafflesia*

(3) *Viscum* (4) *Cuscuta*

191. Each trophic level has a certain mass of living organic material at particular time called as the :-

(1) Nutrient immobilization

(2) Biomagnification

(3) Standing crop

(4) Standing state

192. Consider the following statements and choose the true statements –

(I) Phytoplanktons are consider as primary producer

(II) Zooplanktons are consider as primary consumer

(III) Grasshopper is consider as secondary producer

(IV) Small fishes are consider as primary producer

(V) Wolf is always consider as an example of top carnivore in an ecosystem

(1) I, II, and V (2) I, V and IV

(3) I, II and III (4) I, II, III and IV

193. In an aquatic ecosystem which type of food chain is major conduit for energy flow is

(1) GFC

(2) Parasitic food chain

(3) DFC

(4) Both 1 and 3

194. Which one of the following ecosystem types has the maximum annual net primary productivity?

(1) Tropical deciduous forest

(2) Temperate evergreen forest

(3) Tropical rain forest

(4) Temperate deciduous forest

195. If decomposers are completely removed from ecosystem, the ecosystem functioning is badly affected becasue :-

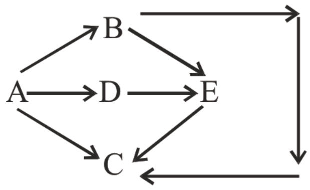
(1) Herbivores will not receive Solar energy

(2) Rate of decomposition of other components will be very high

(3) Mineral movement will be blocked

(4) Energy flow will be blocked

196. Which species is most likely to be decomposer ?



(1) Species A

(2) Species E

(3) Species C

(4) Species D

197. Match the column-I and column-II and choose the correct combination from the option :-

Column-I		Column-II	
(a)	Estuary	(i)	Natural terrestrial ecosytem
(b)	Aquarium	(ii)	Natural aquatic ecosystem
(c)	Cropland	(iii)	Artificial Aquatic ecosystem
(d)	Grassland	(iv)	Artificial Terrestrial ecosystem

(1) a-ii, b-iii, c-iv, d-i

(2) a-ii, b-i, c-iii, d-iv

(3) a-iv, b-ii, c-i, d-iii

(4) a-iii, b-iv, c-ii, d-i

198. Out crossing is an important strategy of animal husbandry because it :-

(1) Exposes harmful recessive genes that are eliminated by selection.

(2) Helps in accumulation of superior genes.

(3) Is useful in producing pure lines of animals.

(4) Is useful in overcoming inbreeding depression.

199. Leghorn is a breed of chicken which is famous for:-

(1) Games and cock fighting

(2) Fat

(3) Eggs

(4) Wings

200. Blue revolution is related to:

(1) Fishes (2) Hen

(3) Cow (4) Bee

Read carefully the following instructions :

1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

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